

Over the wire challenges

Krypton challenges

Krypton Level 0 → Level 1b

Level Information

Host: krypton.labs.overthewire.org

Username: krypton1

SSH Port: 2231

Level Goal: Decode a Base64-encoded string to obtain the password for the next level.

Step-by-Step Solution

1. Understand the Task

You are given a Base64-encoded string:

S1JZUFRPTkITR1JFQVQ=

This string contains the password needed to log in to the next level (krypton1) via SSH.

2. Decode the Base64 String

You used an online Base64 decoder (<https://www.base64decode.org/>). Here's the process in detail:

- **Copy the encoded string:** S1JZUFRPTkITR1JFQVQ=
- **Paste it into the Base64 decoder.**
- **Decode:** The output is KRYPTONISGREAT (all uppercase).

3. Use the Password to Log In

- **Username:** krypton1
- **Password:** KRYPTONISGREAT

Enter the password when prompted. You are now logged in to Level 1.

Decode from Base64 format

Simply enter your data then push the decode button.

```
S1JZUFRPTk1R1JFQVQ=
```

 For encoded binaries (like images, documents, etc.) use the file upload form a little further down on this page.

UTF-8  Source character set.

☐ Decode each line separately (useful for when you have multiple entries).

 Live mode OFF Decodes in real-time as you type or paste (supports only the UTF-8 character set).

 **DECODE**  Decodes your data into the area below.

```
KRYPTONISGREAT
```

Level 1 → Level 2 Writeup

Level Information

- **Host:** krypton.labs.overthewire.org
- **Username:** krypton1
- **SSH Port:** 2231
- **Level Goal:** Find the password for Level 2, which is encrypted using a rotation (Caesar) cipher and stored in the file krypton2. The ciphertext preserves plaintext word boundaries, making it slightly less standard.

Step-by-Step Solution

1. Understand the Challenge

- The file krypton2 contains the password, but it is encrypted with a Caesar (rotation) cipher.
- The clue states that while most ciphers group letters into 5-letter chunks (hiding word boundaries), this ciphertext **keeps the actual word breaks**, helping to spot patterns more easily.

2. Examine the Ciphertext

- Open the file krypton2 after logging in to your account from Level 1.
- You might see ciphertext similar to:
YRRIY GJB CNFFJBEQ EBGGRA

3. Identify the Cipher Used

- Based on the challenge hint ("simple rotation"), this is a classic Caesar cipher.

4. Brute Force the Caesar Cipher

- Go to dcode.org's Caesar Cipher brute-force tool.
- Paste the ciphertext from the file into the tool.
- Run the brute-force decryption. dCode will return all 25 possible shifts at once with likely English results at the top.

5. Analyze the Output

- Review the output. Valid English words should appear in one of the results.
- In the shown result, the decrypted text with a shift of 13 (ROT13) is likely:

LEVEL TWO PASSWORD **ROTTEN**

- The password for Level 2 is thus:
ROTTEN

6. Log in to Level 2

- SSH into Level 2 using the username krypton2 and password **ROTTEN**

Search for a tool

★ [SEARCH A TOOL ON DCODE](#)

★ [BROWSE THE FULL DCODE TOOLS' LIST](#)

Results

Brute-Force mode: the 25 shifts (for the alphabet ABCDEFGHIJKLMNOPQRSTUVWXYZ) are tested and sorted from most probable to least probable.

↑↓	↑↓
→13 (←13)	LEVEL TWO PASSWORD ROTTEN
→24 (←2)	ATKTA ILD EPHLDGS GDIITC
→23 (←3)	BULUB JME FQIIMEHT HEJJUD
→9 (←17)	PIZIP XAS TEWWASVH VSXXIR

CAESAR CIPHER

Cryptography › Substitution Cipher › Caesar Cipher

CAESAR CIPHER DECODER

★ CAESAR SHIFTED CIPHERTEXT

Test all possible shifts (26-letter alphabet A-Z)

► **DECRYPT (BRUTEFORCE)**

MANUAL DECRYPTION AND PARAMETERS

★ SHIFT/KEY (NUMBER):

Ⓢ **USE THE ENGLISH ALPHABET (26 LETTERS FROM A TO Z)**

Krypton Level 2 → Level 3 Writeup

Level Information

- **Host:** krypton.labs.overthewire.org
- **Username:** krypton2 **Pass :** ROTTEN
- **SSH Port:** 2231
- **Level Goal:** Determine the exact Caesar cipher shift used to encode the encrypted key file and decrypt it to find the password for Level 3.

Step-by-Step Solution

2. Navigate to the Challenge Directory

```
cd /krypton/krypton2
```

3. Read the README File

- There is an encrypted file called keyfile.dat.
- A program in the directory can encrypt input using the same Caesar cipher.
- The exact shift value is unknown but you know it's a Caesar cipher (traditionally shift 3, ROT13 is shift 13).
- You must find the exact shift to decrypt the keyfile.

4. Determine the Shift Value

- **Encrypt a test input** and find out the shift from the output.

```
krypton2@krypton:~$ mktemp -d
/tmp/tmp.kGRZpysa1S
krypton2@krypton:~$ cd /tmp/tmp.kGRZpysa1S
krypton2@krypton:/tmp/tmp.kGRZpysa1S$ ls
krypton2@krypton:/tmp/tmp.kGRZpysa1S$ ln -s /krypton/krypton2/keyfile.dat
krypton2@krypton:/tmp/tmp.kGRZpysa1S$ ls
keyfile.dat
krypton2@krypton:/tmp/tmp.kGRZpysa1S$ chmod 777 .
krypton2@krypton:/tmp/tmp.kGRZpysa1S$ ls
keyfile.dat
krypton2@krypton:/tmp/tmp.kGRZpysa1S$ echo "AAA" > example.txt
krypton2@krypton:/tmp/tmp.kGRZpysa1S$ ls
example.txt  keyfile.dat
krypton2@krypton:/tmp/tmp.kGRZpysa1S$ /krypton/krypton2/encrypt example.txt
krypton2@krypton:/tmp/tmp.kGRZpysa1S$ ls
ciphertext  example.txt  keyfile.dat
krypton2@krypton:/tmp/tmp.kGRZpysa1S$ cat ciphertext
MMMMkrypton2@krypton:/tmp/tmp.kGRZpysa1S$ ls
```

- The letter you get is M, meaning the Caesar cipher shifts letters by 12 positions (A to M is the 13th letter, shift 12).

9. Calculate Decryption Shift

- Since encryption shifts forward by 12, decryption shifts backward by 12.

10. Decrypt the Encrypted Password

- You are given the encrypted key: OMQEMDUEQMEK.
- Shift each letter 12 steps backward (or 14 forward) to get the plaintext password.

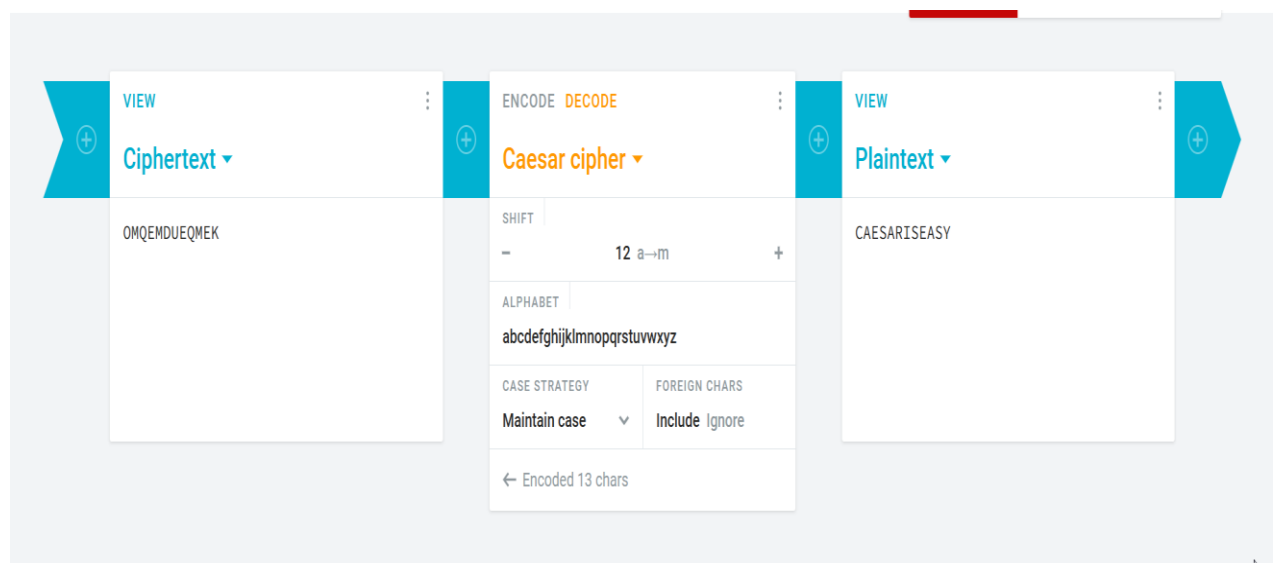
11. Use an Online Tool or Command Line

- Use a tool like [cryptii.com Caesar cipher](https://cryptii.com/caesar-cipher) to quickly decrypt by shifting letters backward by 12.
- Result:

CAESARISEASY

12. Login to Level 3

- Use password CAESARISEASY to SSH as krypton3 on port 2231:



Krypton Level 3 → Level 4 Writeup

Level Information

- **Host:** krypton.labs.overthewire.org
- **Username:** krypton3 **Pass :** CAESARISEASY
- **SSH Port:** 2231
- **Level Goal:** Decrypt a set of intercepted messages, which are encrypted using a monoalphabetic substitution cipher. The goal is to obtain the password for Level 4.

Step-by-Step Solution

1.Navigate to the Challenge Directory

```
cd /krypton/krypton3
```

2. Read the README

- Scan the README file for hints:
- The cipher is not a Caesar cipher, but a more general substitution cipher.
- Multiple ciphertext files are available (found1, found2, etc.).
- All original plaintexts are in English.
- Frequency analysis is suggested as a method for cracking the cipher.

4. Inspect the Ciphertexts

```
krypton3@krypton:~$ cd /krypton/krypton3
krypton3@krypton:~$ ls
found1 found2 found3 HINT1 HINT2 krypton4 README
krypton3@krypton:~$ cat found1
CGZNL YJBEH QYDLQ ZSHQK NZCYD SNQVU BFGBK GQUQZ QSUQW UZCYD SNQDS UDOCKJ ZCYDS NZQSU QNUZB WSRNZ QSUQW UDOCKJ CUBGS BXJDS UCTYV SUJQG WTBUIJ KQMSV LFGBK GGGZL LVYCB GJSZO GCHMS UCJQU QJLYS BXUMA UJCJM JCBGZ CYDSN CGKDC ZDSQZ DVSJJ SNGGJ D
SYVQ CGJSD JCUNS VYQZS NALQV SJJSN UBTSX COSNG HTASN BXYBU CJCBG UMBKG JDSQV YDQAS JXBNS DQTYV SKCJD QUDCK JBYQK BMVNA SNSVY QZSMA LHAKB MYNAS ZBTSS QGMBB BGJDS TSJDB WCUQK TSNQK JSNBN VQMUZ QSUQW KQBMU SMCJJ BZBTT MGCZQ JSKJCJ DDCUE SG
SNQ VUJDS SGEZL YKCBG UJZYV SIKXB TSNAL QZQSU QNZCY DSNCK BXJSG CGZBN YBNQJ SMIQY QNZBX TENSZ BTYVS OUZDS TSUUM ZDUQJ DSICE SONSZ CYDSN QGMBJ CVIDQ UTMWS NGQYV VCEQJ CGGCG JDSNB JUULJ STQJK CJDQV VUCGE VSQYV DQASJ UNALU CJMJC BGECY DSN
UJ DSQZS UQNZC YDSNC USQUC VLANB FSGQG WCGYH QZCZC SRIXS NUSUW SGIQJ VYLBG ZBTIM GCZQJ CBGUS ZMNCJ LUDQF SUYSQ NSYNB WAZSW TBUJB XDCUF GBXKG BNFAS JKSJG QMDC USQNV LYVQL UKSNS TQCGV LZBTS WCSUQ GMDCK JBNCS UESGN SUDSN QCUJW JBJDS YSQF
B XUBYD CUJZC QJCBG QGMBN JCJUN LAJDO SSGWB XJDSU COJSS GJDSZ GJMML GSODJ SKNBJ STQGG VLJNQ ESWKS UMGJC VQABM JCGZY MCGES DQTVS JFCGE VSQNG GHTQZ ASJQZ BGUCW SINSU BJSBK JDSXC GSUJS OQTYV SUGGJ DSSGE VCUDV QGEMQ ESCGD CUVQU JYQQU SDSKN
BJSJN QECZB TSWCS UQVUB FGBKG QUNBT QGZSU QGKZB VYQAB JCJDB JDSNY VQJKN CEDJU TQGLB XDCUY VQJUK SNSYM AVCUO SMCWS WCJCB GUBXJ QNLGC EHMVQ CJLQG WQZZM NQZLN MCGGE DCUVC XSJCT SQGWC GJXBB XDCUX BNTSN JDSQJ NCZQV ZBIVS QEMSU YMAVC
UDSNJ DSKCN UJXBV CBQZB VYSZJ SWSMC JCBGB XDCUW NQTCJ CZXBN FUDQJ JCGZY MWSHQ VYAMJ JKBXK JDSYV QJUGB KNSZB EGCUS WQUOD QFSUY SONSU krypton3@krypton:~$ cat found2
QJZDB MEDGB QJJSQ WQZBS NSZBN WUXBN JDSYS NCBWU MNICI STBUJ ACBEN QYDSN UQENS SJQJQ UDOFS UYSON SKQUS MNQZJ SMCJJ DSFCG EUGSK UZDBB VCGUJ NQJXB MNQXN SSUZD BBVZD QNJSN SMCQG ABMQU HMQNJ SBNXQ TCYSK NBTDC UOBTS ENQTT QNUZO BBVUI QNCSC C
GHMQ VCJLW MNGGE JDSYV CPQAS JDQGS NQAMJ JDSZM MNICM VMTQJ UNICZJ QJSMN LVQKJ DNBMG DBMJS GEVGG WQGUJ DSUZD BBVKB MVMQJ ISYNB ICNSW QGCGJ SGUCI SSIMZ QJCBG CGVQJ CGENQ TTONQ GWJDS ZVQUU CZUQJ JDSQE SBYUD QFSUY SQNST QNCSW NJDSL SQNBV MQ
GGS DQJQD KQJLD SZBBU CUJBN LZBMN JBXJD SMCBZ SUSBX KBNZS UJSSNC UUMSW QTONN CQESV CZSGZ SBGGB ISTAS NJXBB XQJQD QKQLU GSCEP ABMNU YBUJS WABGW UJDSG SOTWQ LQUUM NSJLJ DQJJD SSKNS NSGBC TYSNC TSGJU JBJDS TQNNC QESJD SZBMY VSTQL DQISQ NNQ
GE SNQDS ZSNST BGLCG UBTSQ QJJSU CGZSJ DSKBN ZSUJS NZDQG ZSYVB NQVVB KSNJD STQNN CQESA QGGUJ BASMS QMBGZ SCGUJ SQNBX JDSMU WQJJD NESTC TSUQG GSUYN SEGGQ ZLZBM YMDQI SASGG JDSNS QUBGX BNJDC UUCOT BGJDU QXJSN JDSQJ MNQCE SUDSE QISAC NJDZ
B QNMQE ZSNNU MUQGG QKBKX QJQZY QJUSW BGTOL JKGCU UBQAG TGSQJ GNMOM EDJSN RMCNJ DXBYV BKSQJ VTBUIJ JKBLS QNUVQ JSNQG WKSNS AQVJC USWBG XSANN QMLDQ TGSJN CSWBK MFGGB KQZQN USUQJ JDSQE SBYQS WKQUA MNCW BGOME MUJQX JSNJD SACKJ DBXJD SJKCG
UJDSN SQNBX SKQCU JBNCK QVJNQ ZSYVB UDOFS UYSON SMCJC VQSCU TSGJC BGSNQ UYONJ BXJDS VGBWB GJDSQ JNSUZ SSGCG ASZQM USBXJ DCUEQ YUZDB VQUNW SKNSJ BJDSL SQNUA SJXSS GQGMQ UUDQF SUYSQ NSUWB UJLSQ NUACB ENQYD SNUQJ JSTYJ CGEJB QZBBW GJXBN
JDCUY SMCWQ DQJSN SYBNJ SHTQG LQYBZ NLYDQ WJZBN CSUGC ZDBVQ UNKXS UDOFS UYSON SUVCN UJACB ENQYD SINSZ BMGJS WQJUN QJXBN WJSES GWJQJ JUDQF SUYSQ NSYVS NJDSJ BKCXB NVBGM BGJBS UZQYS YNBUS ZMJCX GXBNN SSNWB QZDQG EQGBJ DSNAC EDJSS GJDSZ G
JMML UJBNL DQJUD QFSUY SONSU JQNZC GEDCU JDSQJ NCZQV ZQNSS NTCGN CGEJD SDBNU SUBXJ DSQJN SYQJN BGUGB VGBWB GRGGB QMANL LNSYB NJSMJ DQJUD QFSUY SONSU QNASS GQZBN GJNLU ZDBBV TQJJS NUBTS JKSJG CSJQZ SGMJN LUZDB VQNUD QJISUM EESUJ SNQJQ JU
DQF SUYSQ NSTQL DQJSA SSGST YVBLB WQUUQ ZDBBV TQJJS NALQV SQDGM SNOBE DJBGB XVQSZ QUDCN SQZQJ DBVCZ VQGBB KGSNK DBQGT SMOZS NJQCG KCVCY QTUQD FSUQJ XJSGG DCUKC VVGBS ICMSH ZSUMA UJQBJ CQJSS UNMZU JBNCS UBJDS NJDQG DSQNU QJZBV VYSZJS WQX
JDS NDCW SQJDK krypton3@krypton:~$ cat krypton4
KSVWV BGSDJ SVSIS VYBMN YQJUK BNICU ANMJS krypton3@krypton:~$
```

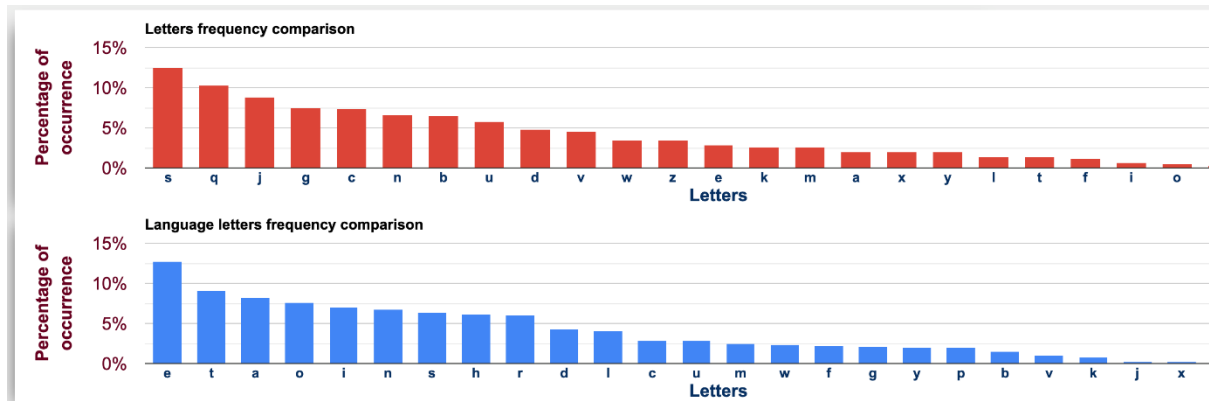
5. Apply Frequency Analysis or Automated Tools

- **Automated Solver:**
Use a tool like **quipqiup** (quipqiup.com):

- Copy the full ciphertext from one of the files (e.g., found1).
- Paste it into the quipqiup solver.

- The tool analyzes the substitutions and produces plaintext output.
- Paste the ciphertext along with the found1 contents

6. Extract the Password



Puzzle:

JULIJ STQIK CJDQV VUCGE VSQVY DQASTJ UMAUJ CJMJC BGZCY DSNUIJ DSZQS UQNZC YDSNC USQUC VIANB FSGQG WCGYN QZJCZ SBXXS NUSUU SGJCQ VVLGB ZBTTH GCZQJ CBGUS ZMNCJ LUDQF
 SUYSQ NSYNB WNZSW TBUJB XDCUF GBKKG BNFAS JKSSG QNGDC USQNV LYVQL UKSNS TQCGV LZBTS WCSUQ GMDCU JBNCB UESGN SUDSN QCUSW JBIDS YSQFB XUBDY CUJZC QJCBG QGMQN JCUJN
 LALJD SSGNB XJDSU COJSS GJDSZ GJMLL GSODJ SKNBZ STQCG VLJNQ ESMCS UMGJC VQABM JCGZV MNCGE DQTVS JFCGE VSQNQ GWTQZ ASJDZ BGUCW SNSMU BJSBX JDSXC GSUJS OQTYV SUCGJ
 DSSGE VCUDV QGEMQ ESCGD CUVQU JYDQU SDSKN BJSDN QECZB TSWCS UQVUB FGBKG QUNBT QGZSU QGWZB VQAB NQJSW KCJDB JDSNY VQLKN CEDJU TQGLB XDCUY VQLUK SNSYM AVCUD SWGCS
 WCJCB GUBXI QNLGC EHMVQ CJLQG WQZHM NQZLW MNCGE DCUVC XSJCT SQGMC GJKBZ XDCUX BNTSN JDSQJ NCZQV ZBVVS QEMSU YMAVC UDSWJ DSXCN UJXBV CBQZB VVSZJ SWSWC JCBGB XDCUW
 NQTQJ CZKBN FUJDQ JCGZV MMSWQ VWAMJ JKBXB JDSYV QLUGB KNSZB EGCUS WQUUD QFSUY SQNSU KSVVV BGSJD SVSIS VXBMM YQUUK BNMCU ANMJS

Clues: For example G-R QWV=THE

Solve

⊗ automatically selected patristocrat mode; you can override by using the drop down menu next to the solve button.

known encryption techniques it is a type of substitution cipher in which each letter in the plain text is replaced by a letter some fixed number of positions down the alphabet for example with a shift of a would be replaced by db would become e and soon the method is named after julius caesar who used it to communicate with his generals the encryption step performed by a caesar cipher is often incorporated as part of more complex schemes such as the vigenre cipher and still has modern application in the rot system as with all single alphabet substitution ciphers the caesar cipher is easily broken and in practice offers essentially no communication security shakespeare produced most of his known work between and his early plays were mainly comedies and histories genre she raised to the peak of sophistication and artistry by the end of the sixteenth century next he wrote mainly tragedies until about including hamlet king lear and macbeth considered some of the finest examples in the english language in his last phase he wrote tragicomedies also known as romances and collaborated with other playwrights many of his plays were published in editions of varying quality and accuracy during his lifetime and in two of his former theatrical colleagues published the first folio a collected edition of his dramatic works that included all but two of the plays now recognised as shakespeare's well done the level four password is brute

1 -3.542 al kryts oz reth ye keiner kathired no glow lene keiner kathir shin hafs kathir keiner komi or keiner hafs noli of shina vtd in selm von swami dy glow lil krytsaol sikh laquinas an esy ti of nub nsas usa ol kathiral whakhi ekhd is sir al sh it deal sic sanri t de kim by edis sir no vifac im luv birofton as aol n mowl shied the bis for ice v tdi washen hafs of e woud mbirit de kim by mb woud mbikovi i e l m n ool shiv is homan levi mef sir judaun keiner who unimas so kovvu lakesi wash hanzil ired nshi il krytsa oln sittir for vim by ekeiner kathiran of silal kortores im enters of

0 -1.395 in cryptography a caesar cipher also known as a caesars cipher the shift cipher caesars code or caesar shift is one of the simplest and most widely known encryption techniques it is a type of substitution cipher in which each letter in the plain text is replaced by a letter some fixed number of positions down the alphabet for example with a shift of a would be replaced by db would become e and soon the method is named after julius caesar who used it to communicate with his generals the encryption step performed by a caesar cipher is often incorporated as part of more complex schemes such as the vigenre cipher and still has modern application in the rot system as with all single alphabet substitution ciphers the caesar cipher is easily broken and in practice offers essentially no communication security shakespeare produced most of his known work between and his early plays were mainly comedies and histories genre she raised to the peak of sophistication and artistry by the end of the sixteenth century next he wrote mainly tragedies until about including hamlet king lear and macbeth considered some of the finest examples in the english language in his last phase he wrote tragicomedies also known as romances and collaborated with other playwrights many of his plays were published in editions of varying quality and accuracy during his lifetime and in two of his former theatrical colleagues published the first folio a collected edition of his dramatic works that included all but two of the plays now recognised as shakespeare's well done the level four password is brute

- In your session, after solving, you observed the plaintext included:

level four password is brute

Krypton Level 4 → Level 5 Writeup

Level Information

- **Host:** krypton.labs.overthewire.org
- **Username:** krypton4 **Pass :** BRUTE
- **SSH Port:** 2231
- **Level Goal:** Decrypt Vigenère-encrypted ciphertext to obtain the password for Level 5. The key length is known (6), and the ciphertext is short, so longer intercepted ciphertexts must be used to find the key.

Navigate to the Challenge Directory

```
cd /krypton/krypton4
```

1. Understand the Challenge

- The README explains the encryption uses a **Vigenère cipher**.
- The key length is 6, which is key information for breaking the cipher.
- The ciphertext flag (HCIKV RJOX) is too short for reliable frequency analysis.
- Use the longer ciphertext files (found1 and found2) to determine the key.

2. Use an Online Tool to Break the Cipher

- Go to [dcode.fr Vigenère cipher tool](https://dcode.fr/Vigenere-cipher-tool).
- Paste the longer ciphertext from found1.
- Set the key length to 6.
- Click Decrypt to analyze the ciphertext using the known key length.

3. Identify the Key

- The tool proposes the key:
FREKEY
- This key produces readable plaintext output, so it is likely the correct key.

4. Decrypt the Flag Ciphertext

- Open the decryption tab on the tool.
- Give the key FREKEY.
- Input the short ciphertext flag:

HCIKV RJOX

- The decrypted result is:

CLEARTEXT

- This is the password for Level 5.

11

11

THESO LDIER WITHT HEGRE ENWHI
 SKERS LEDTH EMTHR OUGHT HESTR
 EETSO FTHEE MERAL DCITY UNTIL
 THEYR EACHE DOTHER OOMWH ERETH
 EGUAR DIANO FTHEG ATESL IVEDT
 HISOF FICER UNLOC KEDTH EIRSP
 ECTAC LESTO PUTTH EMBAC KINHI
 SGREA TBOXA NDTHE NHEPO LITEL
 YOPEN EDTHE GATEF OROUR FRIEN
 DSWHI CHROA DLEAD STOTH EWICK
 EDWIT CHOFT HEWES TASKE DDORO
 THYTH EREIS NOROA DANSW EREDT
 HEGUA RDIAN OFTHE GATES NOONE
 EVERW ISHES TOGOT HATWA YHOWT
 HENAR EWETO FINDH ERINQ UIRED
 THEGI RLTHA TWILL BEEAS YREPL
 IEDTH EMANF ORWHE NSHEK NOWSY
 OUAERE INTHE COUNT RYOFT HEWIN
 KIESS HEWIL LFIND YOUAN DMAKE
 YOUAL LHERS LAVES PERHA PSNOT
 SAITD HESCA RECRO WFORW EMEAN
 TODES TROYH EROHT HATIS DIFFE
 RENTS AIDTH EGUAR DIANO FTHEG
 ATESN OONEH ASEH RDEST ROYED
 HERBE FORES OINAT URALL YTHOU
 GHTSH EWOL DMAKE SLAVE SOFYO
 UASSH EHASO FOTHER ESTBU TTAKE
 CAREF ORSHE ISWIC KEDAN DFIER
 CEAND MAYNO TALLO WYOUT ODEST
 ROYHE RKEEP TOTHE WESTW HERET
 HESUN SETSA NDYOU CANNO TFAIL
 TOFIN DHERT HEYTH ANKED HIMAN
 DBADE HINGO ODBYE ANDTU RNEDT
 OWARD THEWE STWAL KINGO VERFI
 ELDSO FSOFT GRASS DOTTE DHERE

★ ALPHABET

ABCDEFGHIJKLMNOPQRSTUVWXYZ

×

▶ AUTOMATIC DECRYPTION

DECRYPTION METHOD

☐ KNOWING THE KEY/PASSWORD: KEY
 ☐ KNOWING THE KEY-LENGTH/SIZE, NUMBER OF LETTERS: 6
 ☐ KNOWING ONLY A PARTIAL KEY (JOKER=?): KE?
 ☐ KNOWING A PLAINTEXT WORD: CODE
 ☐ VIGENERE CRYPTANALYSIS (KASISKI'S TEST)

★ SHOW VIGENÈRE'S SQUARE/GRID (TABULA RECTA) ☐

▶ DECRYPT

See also: Autoclave Cipher – Beaufort Cipher – Caesar Cipher

VIGENERE ENCODER

★ VIGENERE PLAIN TEXT

dCode Vigenere automatically

★ CIPHER KEY

FREKEY

×

★ ALPHABET

ABCDEFGHIJKLMNOPQRSTUVWXYZ

×

★ PRESERVE PUNCTUATION, LOWERCASE ETC. ☒

★ SHOW VIGENÈRE'S SQUARE/GRID (TABULA RECTA) ☐

▶ ENCRYPT

See also: Beaufort Cipher – Autoclave Cipher – Caesar Cipher

Answers to Questions (FAQ)

What is the Vigenere cipher? (Definition)

The **Vigenère cipher** (sometimes written Viginere) is a polyalphabetic encryption method using a keyword to encode a message.

Invented by the French cryptologist Blaise de **Vigenère** in the 16th century, it is based on the use of a grid/table called a **Vigenère square**

VIEW

+

Ciphertext ▾

HCIVK RJOX

ENCODE DECODE

+

Vigenère cipher ▾

VARIANT

Standard Vigenère cipher ▾

KEY

FREKEY

KEY MODE

Repeat ▾

ALPHABET

abcdefghijklmnopqrstuvwxyz

CASE STRATEGY

Maintain case ▾

FOREIGN CHARS

Include Ignore

→ Decoded 10 chars

VIEW

+

Plaintext ▾

CLEAR TEXT

Krypton Level 5 → Level 6 Writeup

Level Information

- **Host:** krypton.labs.overthewire.org
 - **Username:** krypton5 **Pass :** CLEARTXT
 - **SSH Port:** 2231
 - **Level Goal:** Decrypt a polyalphabetic cipher (Vigenère) where the key length is unknown. Obtain the password for Level 6.
-

Step-by-Step Solution

Challenge Directory

```
cd /krypton/krypton5
```

1. Read the README

- Informs you that the key length is unknown this time.
- You'll need to figure out both the key length and the key—this is now a textbook **Vigenère challenge**.
- Multiple “found” ciphertexts are given for analysis.

2. Crack Key Length and Key Using dCode

- Go to [dcode.fr's Vigenère Cipher Solver](https://dcode.fr/vigenere-cipher-solver).
- Paste the contents of found1 (or a long ciphertext) into the tool.
- Hit **Automatic Decryption** (dCode will try to determine both the key length and the key itself).
- The dCode tool determines:
 - **Key: KEYLENGTH**

3. Decrypt the Level Password Ciphertext

- Use the ciphertext provided for the flag:
BELOS Z
- Using the key KEYLENGTH, decrypt this using the dCode or a similar Vigenère decryption tool.
- **Decrypted output:**
RANDOM

Search for a tool

SEARCH A TOOL ON DCODE

e.g. type 'sudoku'

BROWSE THE FULL DCODE TOOLS' LIST

Results

Warning Showing most likely results

ABCDEFGHIJKLMNOPQRSTUVWXYZ (26)

↑↓	↑↓
NPYPO	GBLUJ
OVDBG	CTXXE
VLWFN	
OXBKT	HPBZA
BRAUX	PJUZM
MACVA	
BZDBX	ZQBEG
QHLVD	GPCRI
MYTKN	
XOUHL	LXEVL
WFNOX	JLQRD
BZIXQ	
EGUEG	QHLVD
GTLBW	OHKEP
RNRXB	
ENPAX	PJUZM
MAUTW	FIUHK
HKVDG	
CAPFO	VWAFY
HLTJQ	UZNLK
GJOUX	
PJUZM	MAUEN
VHNHK	DQEAU
NDTXP	
JTSVH	AXWKH
SAFJH	BWSGW
WQYCX	
WUVWE	AAIXK
KTTQF	QLAFZ
PDPUC	
UZGAH	DNROZ
PJSGG	ANFSZ
TEPVZ	
VLLVY	PQWAN
PLGBA	YTNRU
SELKK	
CCQCY	XHPVW
ABNOX	WSCNE
AMOHV	
PVWAC	YYBTZ
YPOFI	MTWHK
ZAGBL	
IWAUT	JGJLK
NKFID	NNZHR
AQUEG	
MUHNO	KTOGU
BMMKT	XPVYZ
BSOKH	
PRXVG	NPUQA
VHNKJ	YGXRR
XMHWK	
QDZBL	MHWAX
XHVHA	AJOUE
AEFHM	
NRGSA	TLLXT
BEDIC	UYBKK
PDJYS	
AAJNG	LAEYH
DNJIL	EGBHE
FNITF	
NQHGI	WSJAR
HDBYD	CEHNC
UYFYG	
DJGBL	MMNQC
ABZLG	LHCCZ
GBLKJ	
SGGAN	EPGLS
KIDNF	HKLAL
PSNHK	
TVQGT	JJCAA
FBCXN	SUJXT
JVWAG	
BYHSA	QUBEU
UVJEP	QKGBJ
HZJVG	

VIGENERE CIPHER

Cryptography › Poly-Alphabetic Cipher › Vigenere Cipher

VIGENERE DECODER

VIGENERE CIPHERTEXT

KPGEC EGIWO HMFVG NXPBW KPVR SXCCE PWVTM OOIYC XURRV
BHCCS SKOLX XQSEQ RTAOP WNSZK MVDLC PRTRB ZRGPZ AAGGK
ZIMAP RLKVV EAZRT XXZCS DMVVZ BZRW S MNRIM ZSRYP IEOVH
GLGNL FZKHK KCESE KEHDI FLZRV KVFIB XSEKB TZSPE EAZMV
DLCSY ZGGYK GCELN TTUIG MXQHT BJKXG ZRFEX ABIAP MIKWA
RVMFK UGGFY JRSIP NBJUI LDSSZ ALMSA VPNTX IBSMO

PLAINTEXT LANGUAGE

English

ALPHABET

ABCDEFGHIJKLMNOPQRSTUVWXYZ

AUTOMATIC DECRYPTION

DECRYPTION METHOD

KNOWING THE KEY/PASSWORD: KEY

KNOWING THE KEY-LENGTH/SIZE, NUMBER OF LETTERS: 3

KNOWING ONLY A PARTIAL KEY (JOKER=?): KE?

KNOWING A PLAINTEXT WORD: CODE

VIGENERE CRYPTANALYSIS (KASISKI'S TEST)

SHOW VIGENÈRE'S SQUARE/GRID (TABULA RECTA)

DECRYPT

See also: Autoclave Cipher – Beaufort Cipher – Caesar Cipher

VIGENERE ENCODER

VIGENERE PLAIN TEXT

dCode Vigenere automatically

CIPHER KEY

FREKEY

ALPHABET

ABCDEFGHIJKLMNOPQRSTUVWXYZ

PRESERVE PUNCTUATION, LOWERCASE ETC.

SHOW VIGENÈRE'S SQUARE/GRID (TABULA RECTA)

VIEW

Ciphertext

BELOSZ

ENCODE DECODE

Vigenère cipher

VARIANT

Standard Vigenère cipher

KEY

KEYLENGTH

KEY MODE

Repeat

ALPHABET

abcdefghijklmnopqrstuvwxyz

CASE STRATEGY

FOREIGN CHARS

Maintain case

Include Ignore

← Encoded 6 chars

VIEW

Plaintext

RANDOM

Krypton Level 6 → Level 7 Writeup

Level Information

- **Host:** krypton.labs.overthewire.org
- **Username:** krypton6 **Pass :** RANDOM
- **SSH Port:** 2231
- **Level Goal:** Break a periodic stream cipher by exploiting its repeating keystream via a chosen-plaintext attack.

Step-by-Step Solution

Explore the README and Hints

- The challenge uses a stream cipher with a repeating keystream (not a secure one-time pad).
- The hint suggests: "The 'random' generator has a limited number of bits, and is periodic... There is a pattern!"

2. Discover the Keystream

```
krypton6@krypton:~$ cd /krypton/krypton6
krypton6@krypton:/krypton/krypton6$ ls
encrypt6  HINT1  HINT2  keyfile.dat  krypton7  onetime  README
krypton6@krypton:/krypton/krypton6$ cat krypton7
PNUKLYLWRQKGKBEkrypton6@krypton:/krypton/krypton6$ cat HINT1
The 'random' generator has a limited number of bits, and is periodic.
Entropy analysis and a good look at the bytes in a hex editor will help.

There is a pattern!
krypton6@krypton:/krypton/krypton6$ cat HINT2
8 bit LFSR
krypton6@krypton:/krypton/krypton6$ python3 -c "print('A'*100)" > /tmp/plain.txt
krypton6@krypton:/krypton/krypton6$ ./encrypt6 /tmp/plain.txt /tmp/cipher.txt
krypton6@krypton:/krypton/krypton6$ cat /tmp/cipher.txt
EICTDGVIVYZKTHNSIRFXYPFUEOCKRNEICTDGVIVYZKTHNSIRFXYPFUEOCKRNEICTDGVIVYZkrypton6@krypton:/krypton/krypton6$
```

- Through chosen-plaintext attacks, you can recover the keystream, which repeats every 30 characters.
- You create files of 30 A's and 30 B's and encrypt them:

[illegible]

4. Analyze the Flag Ciphertext

- The flag ciphertext given is:

PNUKLYLWRQKGKBE

- To decrypt, for each position, compare the ciphertext letter to the corresponding letter in the known keystream.

5. Python to Decrypt (Automated Solution)

Insert the recovered keystream and ciphertext into the following script:

```
crypt = 'EICTDGYIYZKTHNSIRFXYPFUEOCKRN'
ciphertext = "PNUKLYLWRQKGKBE"
for i in range(len(ciphertext)):
    val = ord(ciphertext[i]) - ord(crypt[i])
    if val < 0: val += 26
    val += ord('A')
    print(chr(val), end='')
print()
```

- This script reveals the plaintext flag.

6. Result

- Running the script prints:

LFSRISNOTRANDOM

- This is the password for Level 7.

Krypton Level 7 Writeup

Level Information

- **Host:** krypton.labs.overthewire.org
- **Username:** krypton7 **Pass:** LFSRISNOTRANDOM
- **SSH Port:** 2231

Message : Congrats you broke Krypton !